

Problem Statement No. 14

AI Agent for Smart Farming Advice

Project: FarmingAI

Domain: Smart Agriculture / Agentic AI

Challenge

Farmers often need timely guidance across multiple areas of crop production and farm management. The project challenge is to provide an integrated AI-based assistant that can combine crop advisory, weather and irrigation information, pest and disease support, market analysis, and agricultural knowledge within a single farmer-facing application.

Objective

Develop an AI agent for smart farming advice that accepts farmer and farming context and produces practical, integrated guidance through a Streamlit application using Python and Groq-based AI models.

Proposed Agentic Solution

- **Crop Advisory Agent:** crop/seed recommendations, fertilizer guidance, and crop schedules.
- **Weather & Irrigation Agent:** current weather, forecast information, and irrigation recommendations.
- **Pest & Disease Detection Agent:** multimodal crop/leaf diagnosis with a text-based fallback.
- **Market Insights Agent:** MSP/cost analysis and break-even calculations.
- **Agricultural Knowledge / RAG Agent:** retrieval from a curated 12-topic agricultural knowledge base.

Application

The agents are coordinated by a central FarmingOrchestrator and presented through a Streamlit dashboard. The application also uses OpenWeatherMap for weather data and Plotly for visualizations.

Core Technology

Python, Streamlit, Groq API and Llama models, OpenWeatherMap API, Plotly, Pandas/NumPy, Pillow, and python-dotenv.

Expected Outcome

A unified AI-assisted farming dashboard that brings multiple advisory functions together so that a farmer can obtain contextual crop, weather, irrigation, pest/disease, market, and agricultural-knowledge guidance from one application.

This document summarizes the Problem Statement No. 14 content represented in the FarmingAI project presentation. It is intended as the project repository copy of the problem statement.